

TATEYAMA AB, Japan

WMO: 476880

Lat: 34.987N

Lon: 139.832E

Elev: 20

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.1	(c) 34.2	(d) 13.9	(e) 11.1	(f) 40.7	(g) 16.1	(h) 12.4	(i) 41.4	(j) 37.3	(k) 48.2	(l) 33.3	(m) 48.4	(n) 5.1	(o) 130	(p) 0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.1	(c) 88.1	(d) 79.3	(e) 86.5	(f) 78.8	(g) 85.7	(h) 78.5	(i) 80.7	(j) 85.7	(k) 79.9	(l) 84.6	(m) 79.2	(n) 83.5	(o) 10.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.2	83.9	78.7	149.3	83.6	77.4	142.4	82.3	44.4	86.4	43.6	85.0	42.8	83.6	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.9	24.3	20.7		DB	28.3	91.8	1.5	1.9	27.2	93.2	26.3	94.3	25.5	95.4	24.4	96.8
				WB	24.8	82.1	1.4	1.0	23.8	82.8	23.0	83.4	22.3	84.0	21.2	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	44.5	45.3	50.3	58.1	65.5	70.9	77.9	80.8	75.2	66.5	57.5	49.5
	DBStd	13.10	4.28	4.86	5.40	5.46	4.01	3.66	3.99	3.30	4.85	4.71	5.05	4.78
	HDD50	472	181	150	64	6	0	0	0	0	0	0	4	67
	HDD65	2653	635	553	457	213	43	2	0	1	37	231	481	
	CDD50	4823	11	17	72	248	480	628	866	954	757	511	228	51
	CDD65	1530	0	0	0	6	57	181	401	489	308	83	5	0
	CDH74	10859	0	0	0	2	72	505	3079	4815	2142	242	2	0
	CDH80	2634	0	0	0	0	1	31	714	1436	439	13	0	0
(14) Wind	WSAvg	9.0	10.5	10.0	10.4	9.7	8.6	8.0	8.5	8.2	7.9	8.2	8.5	10.0
(15) Precipitation	PrecAvg	75.5	3.5	3.9	6.4	6.5	6.7	8.7	6.9	4.8	9.0	9.3	6.1	3.5
	PrecMax	93.8	9.0	7.4	11.5	12.6	12.0	16.7	18.0	13.6	18.2	25.4	18.6	10.2
	PrecMin	49.9	0.0	0.7	2.2	3.3	3.3	3.5	0.2	0.1	2.7	2.0	1.7	0.2
	PrecStd	11.9	2.2	1.9	2.3	2.4	2.2	3.5	4.8	3.6	4.0	5.6	3.7	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	64.6	67.8	73.2	78.6	83.6	89.8	91.1	88.1	82.0	73.5	68.3
		MCWB	57.6	58.9	59.4	64.5	68.0	75.2	78.9	80.2	78.4	73.2	65.7	63.2
	2%	DB	59.0	61.1	64.7	70.2	75.5	80.6	87.7	89.3	86.1	78.8	71.3	64.6
		MCWB	52.5	55.7	58.3	63.2	67.4	73.3	79.2	79.5	78.3	72.3	64.8	58.0
	5%	DB	55.7	57.9	62.8	68.4	73.8	78.7	86.0	87.8	84.3	76.9	69.4	61.2
		MCWB	48.1	51.6	56.9	61.6	66.2	72.3	78.4	79.5	77.2	70.3	62.8	53.6
	10%	DB	53.5	55.2	60.8	66.6	73.0	77.0	84.2	86.2	82.5	74.9	66.6	58.9
		MCWB	45.6	48.1	54.4	60.1	65.8	71.6	77.5	78.9	76.2	68.3	59.9	51.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	60.3	63.1	68.8	72.6	77.4	81.0	82.1	80.8	76.5	69.0	64.9
		MCDB	61.9	63.2	65.3	70.6	75.0	81.3	86.1	86.7	84.8	79.0	71.7	67.5
	2%	WB	53.7	56.8	60.2	65.4	70.4	75.7	80.0	81.0	79.6	73.9	65.8	59.2
		MCDB	58.1	60.6	63.7	69.0	73.6	78.8	85.1	86.0	83.5	77.3	69.4	63.1
	5%	WB	49.7	52.5	58.0	63.5	68.8	74.2	79.2	80.3	78.6	71.3	63.5	55.1
		MCDB	54.7	57.2	62.2	67.6	72.5	77.3	83.9	85.3	82.5	75.6	67.7	60.2
	10%	WB	46.3	49.0	55.3	61.6	67.2	72.6	78.4	79.7	77.7	69.1	61.4	51.9
		MCDB	51.7	53.8	59.8	65.8	71.0	75.9	82.6	84.3	81.6	73.4	65.8	57.2
(35) Mean Daily Temperature Range	MDBR		13.9	13.7	13.5	13.4	12.1	9.8	9.6	10.1	10.8	11.6	13.0	13.5
	5% DB	MCDBR	16.5	16.5	15.1	14.3	14.0	11.6	11.6	11.4	11.4	12.9	14.9	16.1
		MCWBR	13.4	14.2	12.2	9.6	8.2	6.5	5.6	5.0	6.1	8.9	11.0	12.8
	5% WB	MCDBR	15.5	15.5	13.7	11.7	11.1	9.0	10.3	10.1	10.1	11.2	12.5	15.1
		MCWBR	13.8	14.5	13.1	9.6	8.2	6.1	5.5	5.1	6.2	9.3	10.9	13.6
(40) Clear-Sky Solar Irradiance	taub		0.388	0.438	0.500	0.543	0.557	0.573	0.568	0.551	0.500	0.483	0.432	0.389
	taud		2.234	2.091	1.942	1.882	1.910	1.947	2.003	2.044	2.158	2.120	2.179	2.238
	Ebn at Noon		251	250	244	241	239	235	235	236	242	233	233	242
	Edh at Noon		35	44	56	62	62	59	56	53	45	43	36	33
(44) All-Sky Solar Radiation	RadAvg		834	989	1261	1492	1626	1428	1640	1663	1277	958	777	740
(45)	RadStd		67	110	89	156	171	141	266	140	110	100	70	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page